

SMARTX: Porewater SO₄/Cl

July 2026 Samples 333-463

2026-10-08

Contents

0.1	Run Information	2
0.2	Assess Standard Curves	3
0.3	Assess Check Standards	4
0.4	Assess Blanks	5
0.5	Assess Duplicates	7
0.6	Calculate mmol/L concentrations & salinity, add dilutions	8
0.7	Assess Analytical Spikes	9
0.8	Check if samples within the range of the standard curve	9
0.9	Check to see if samples run match metadata & merge info	10
0.10	Visualize Data - Cl	10
0.11	Export Processed Data	18

##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-10-06" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "July"
Site = "GCREW"
run_notes = "" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/SMARTX_202607_333-463_C1.txt"
raw_file_name_so4 = "Raw Data/SMARTX_202607_333-463_S04.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCReW_SMARTX_PW_Processed_C1_S04_202607_333-463.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCReW", "GCReW_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = here("Porewater QAQC Logs", "GCReW_S04_C1_QAQC_Log.csv")
```

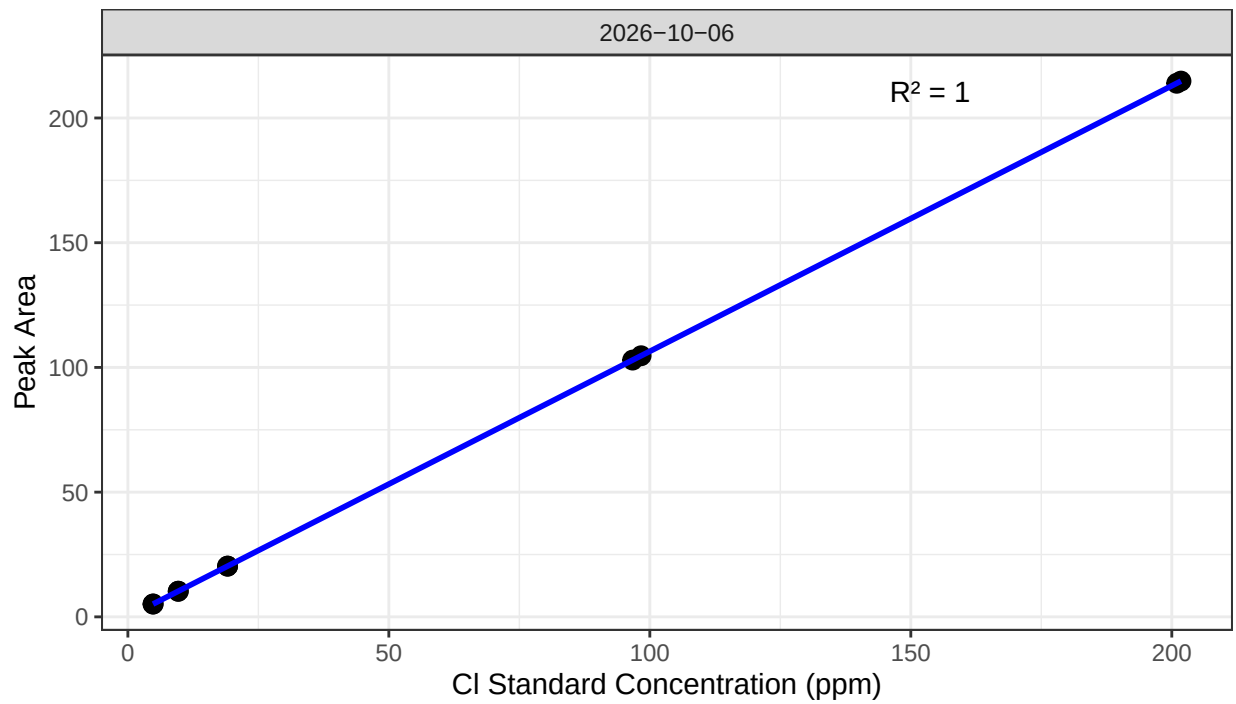
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

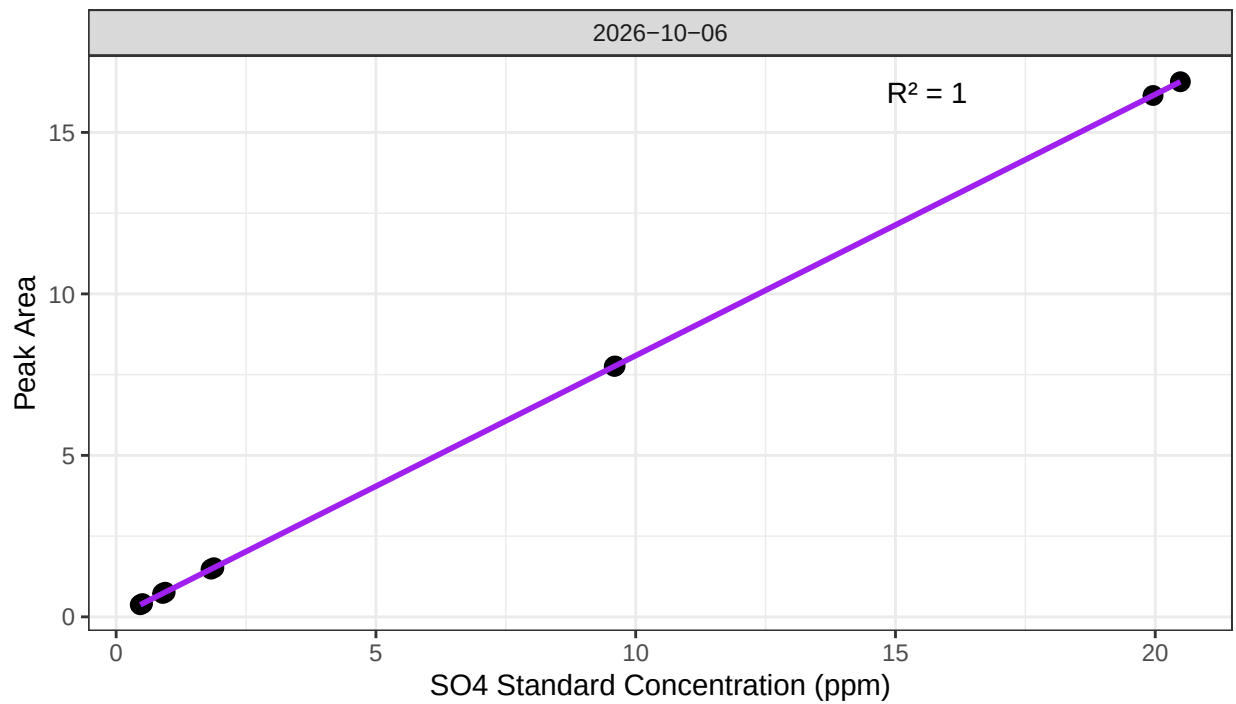
##Import Sample Data

0.2 Assess Standard Curves

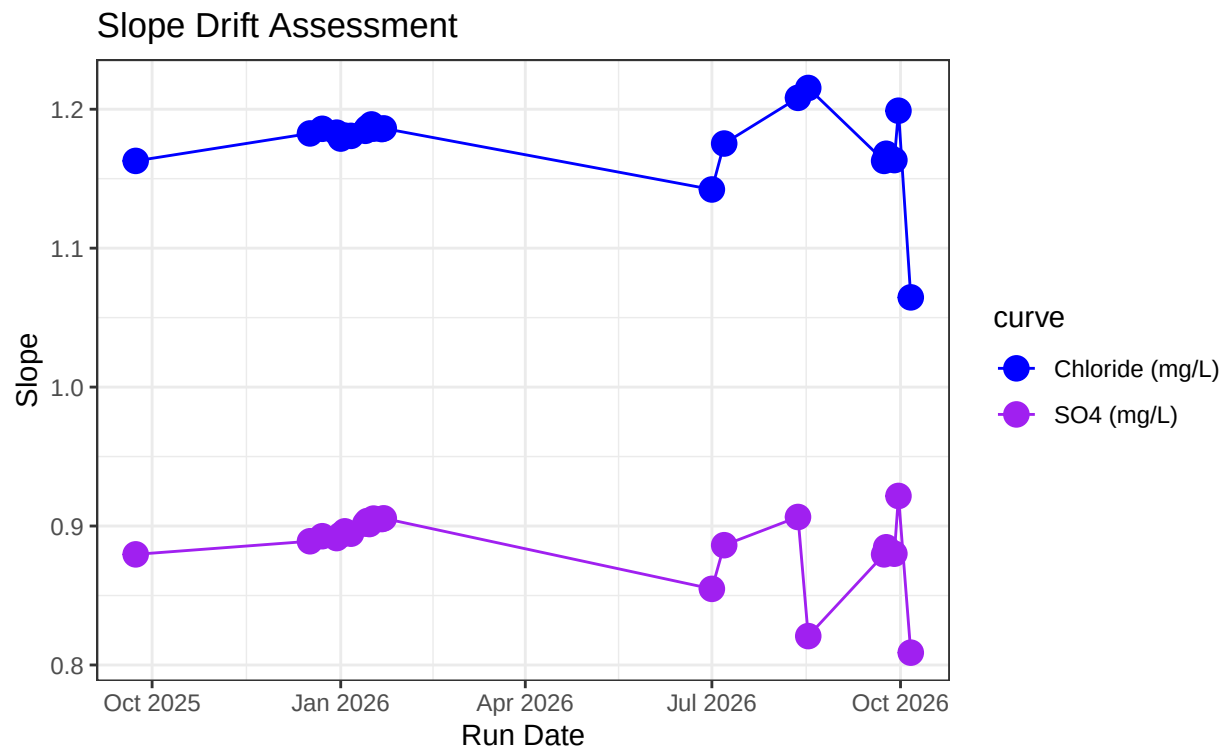
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

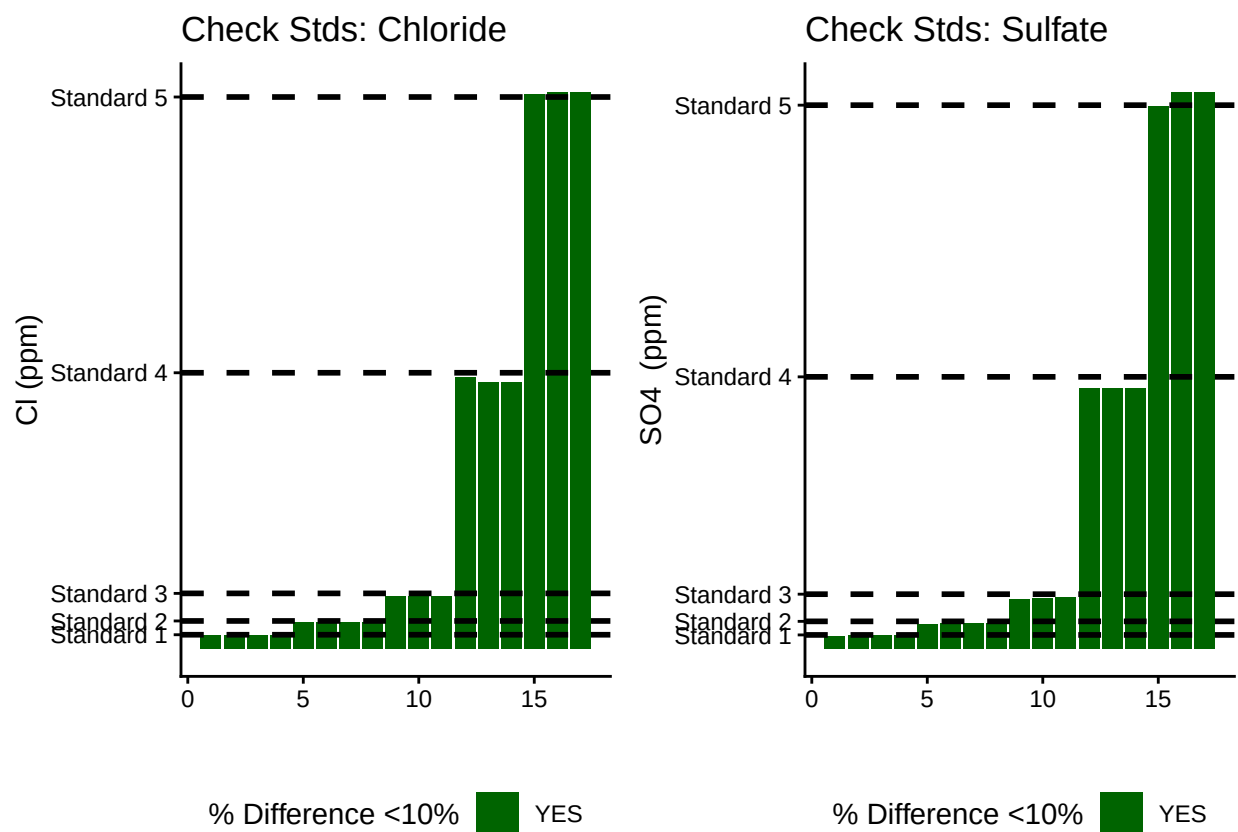
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 4.85 0.0199 0.00411 Chloride Check Standard RSD within Range~
## 2 Standard 2 9.66 0.0161 0.00167 Chloride Check Standard RSD within Range~
## 3 Standard 3 19.1 0.00147 0.0000772 Chloride Check Standard RSD within Range~
## 4 Standard 4 97.2 0.946 0.00973 Chloride Check Standard RSD within Range~
## 5 Standard 5 202. 0.541 0.00268 Chloride Check Standard RSD within Range~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 0.496 0.0209 0.0422 Sulfate Check Standard RSD within Range ~
## 2 Standard 2 0.927 0.0207 0.0223 Sulfate Check Standard RSD within Range ~
## 3 Standard 3 1.86 0.0280 0.0151 Sulfate Check Standard RSD within Range ~
## 4 Standard 4 9.60 0.00764 0.000796 Sulfate Check Standard RSD within Range ~
## 5 Standard 5 20.3 0.308 0.0152 Sulfate Check Standard RSD within Range ~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



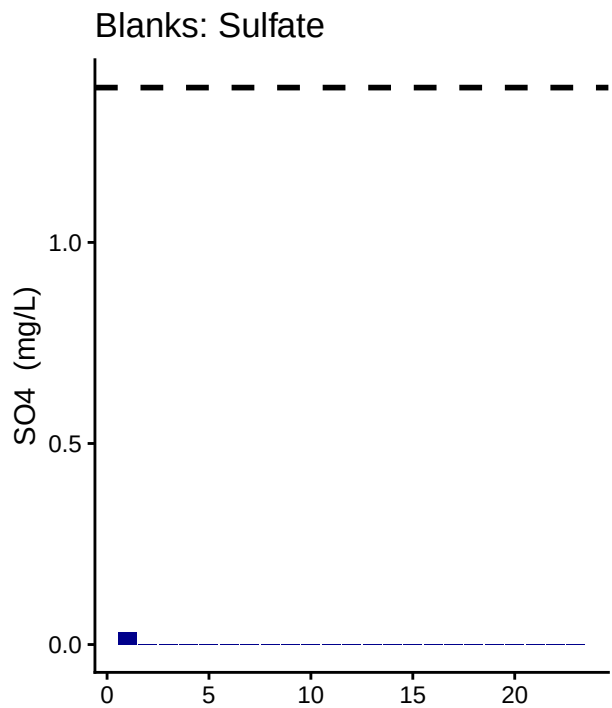
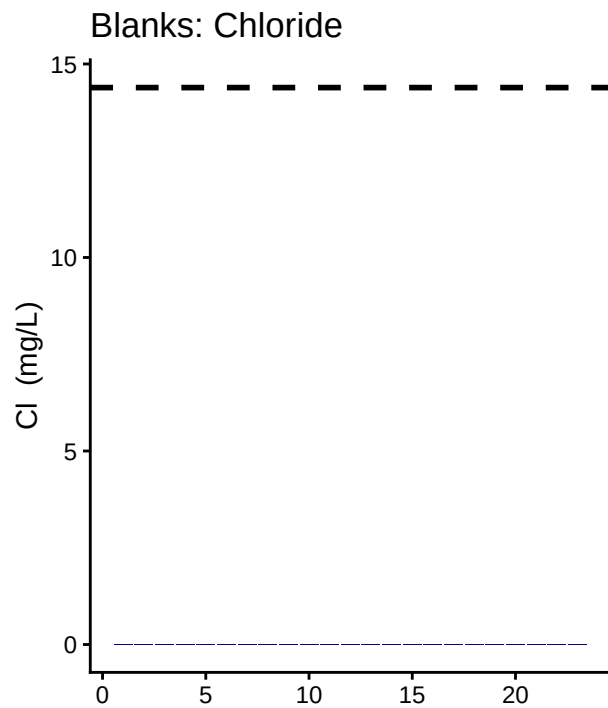
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



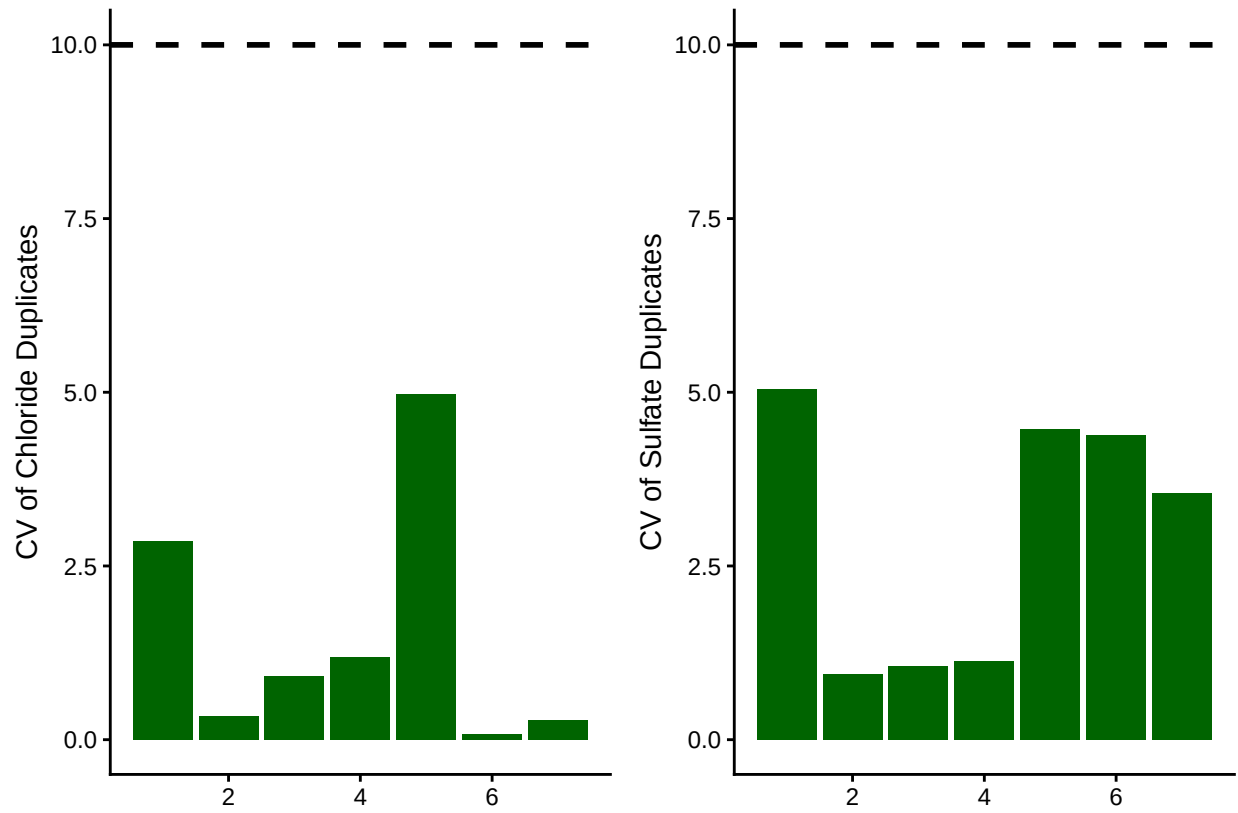
```
## Chloride blanks mean ppm:
```

```
## [1] 0.003773913
```

```
## Sulfate blanks mean ppm:
```

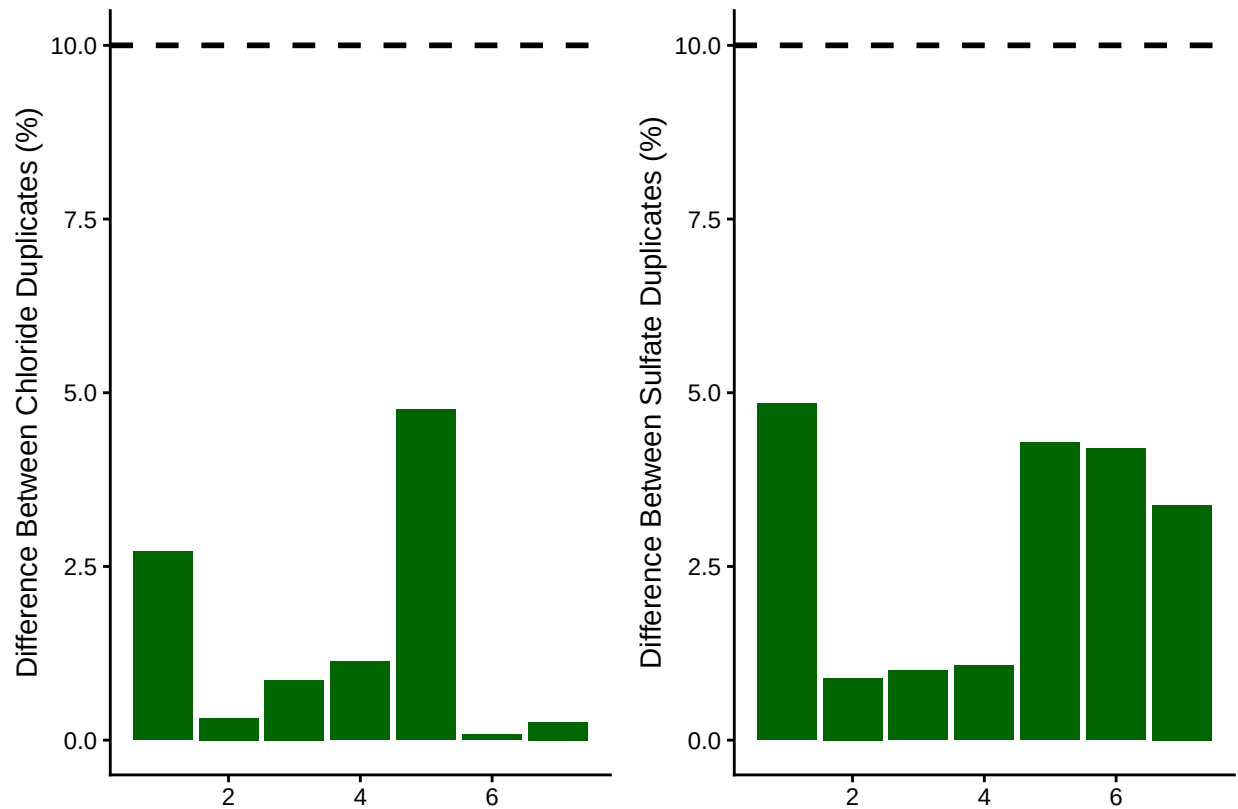
```
## [1] 0.001317391
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

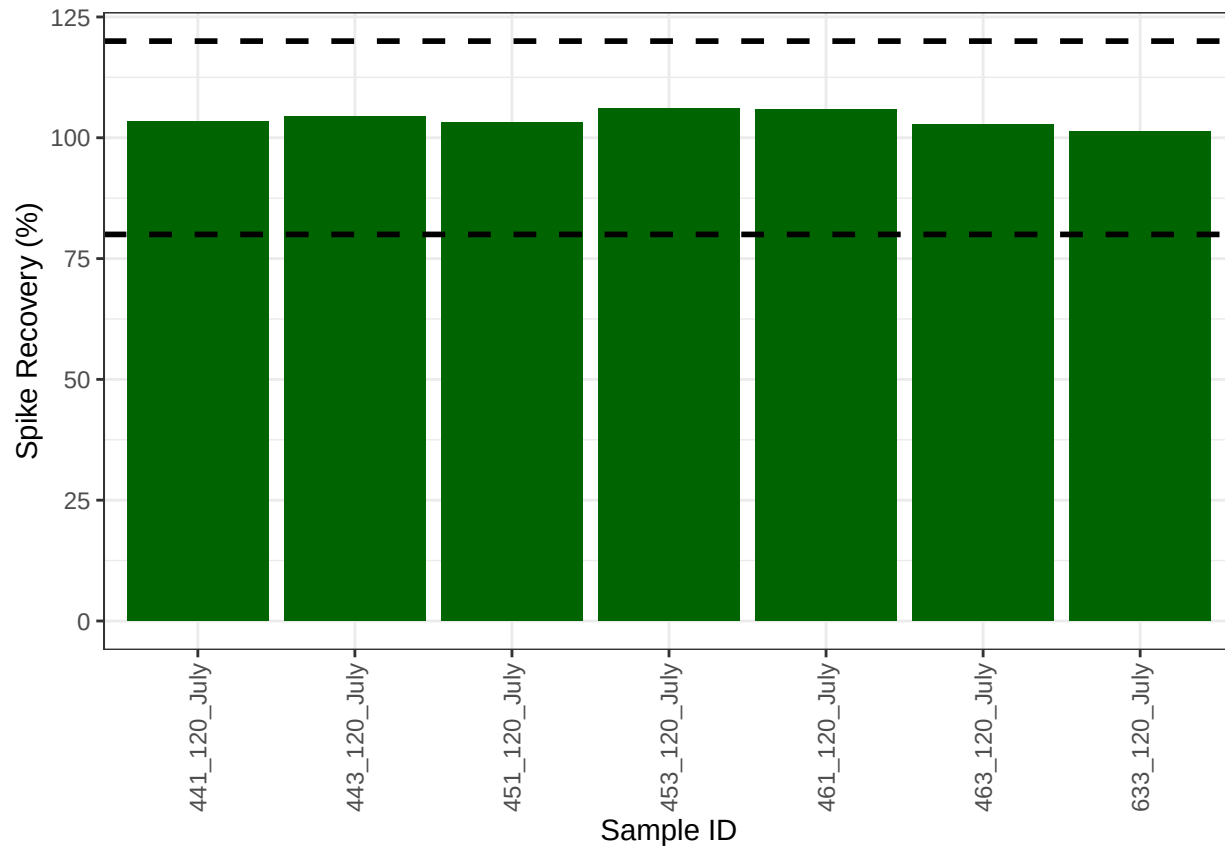
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
all_dat$Dilution <- Dilution
# head(all_dat)
```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

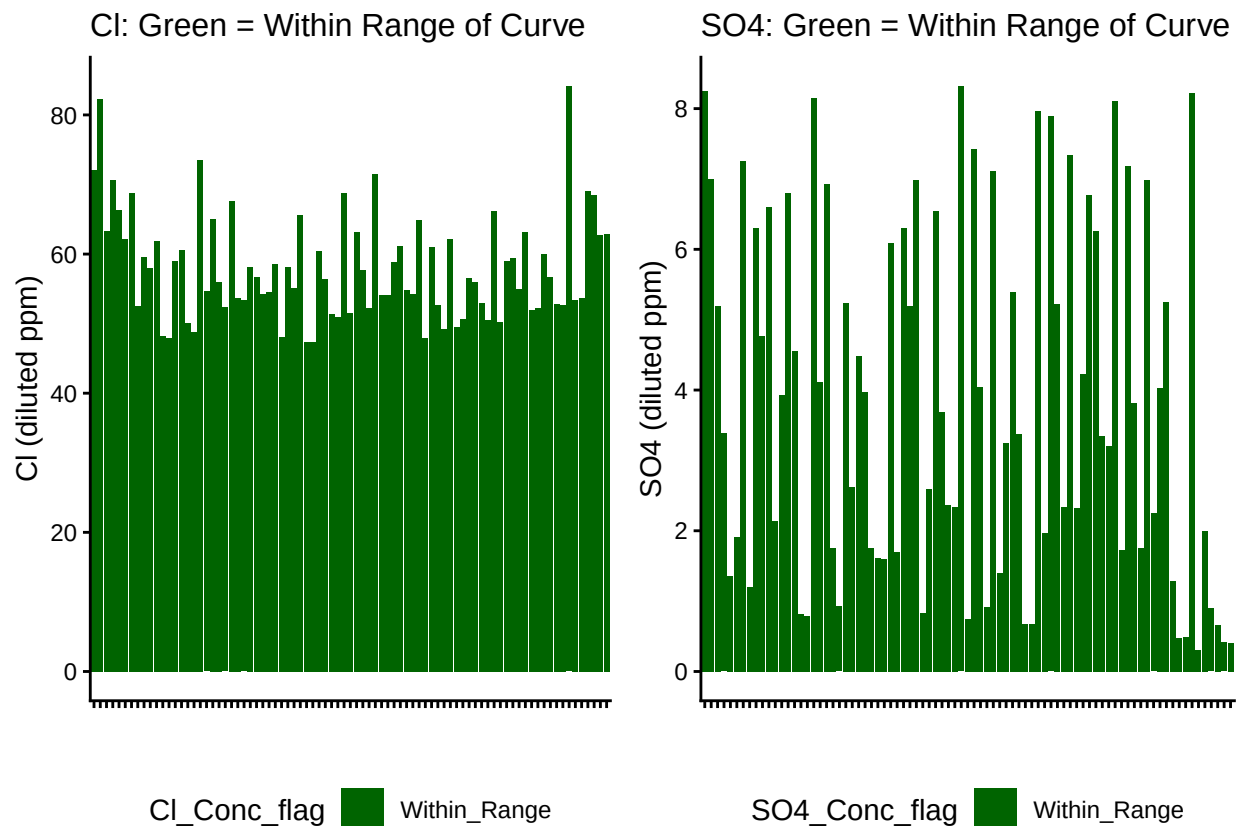


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

0.9 Check to see if samples run match metadata & merge info

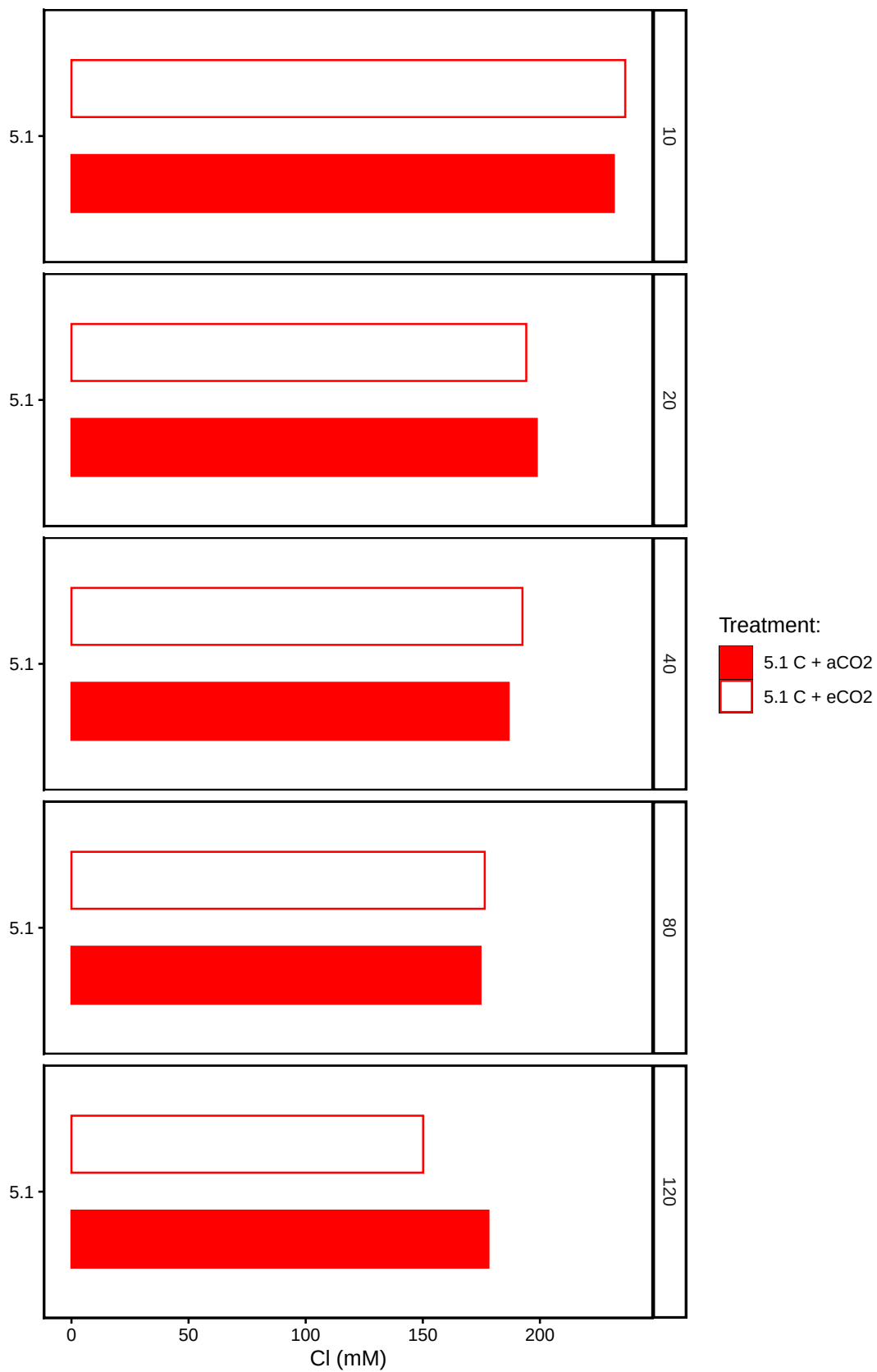
```
## Some sample IDs are missing from metadata.
```

```
## [1] "310_10_July_rerun_2026"
```

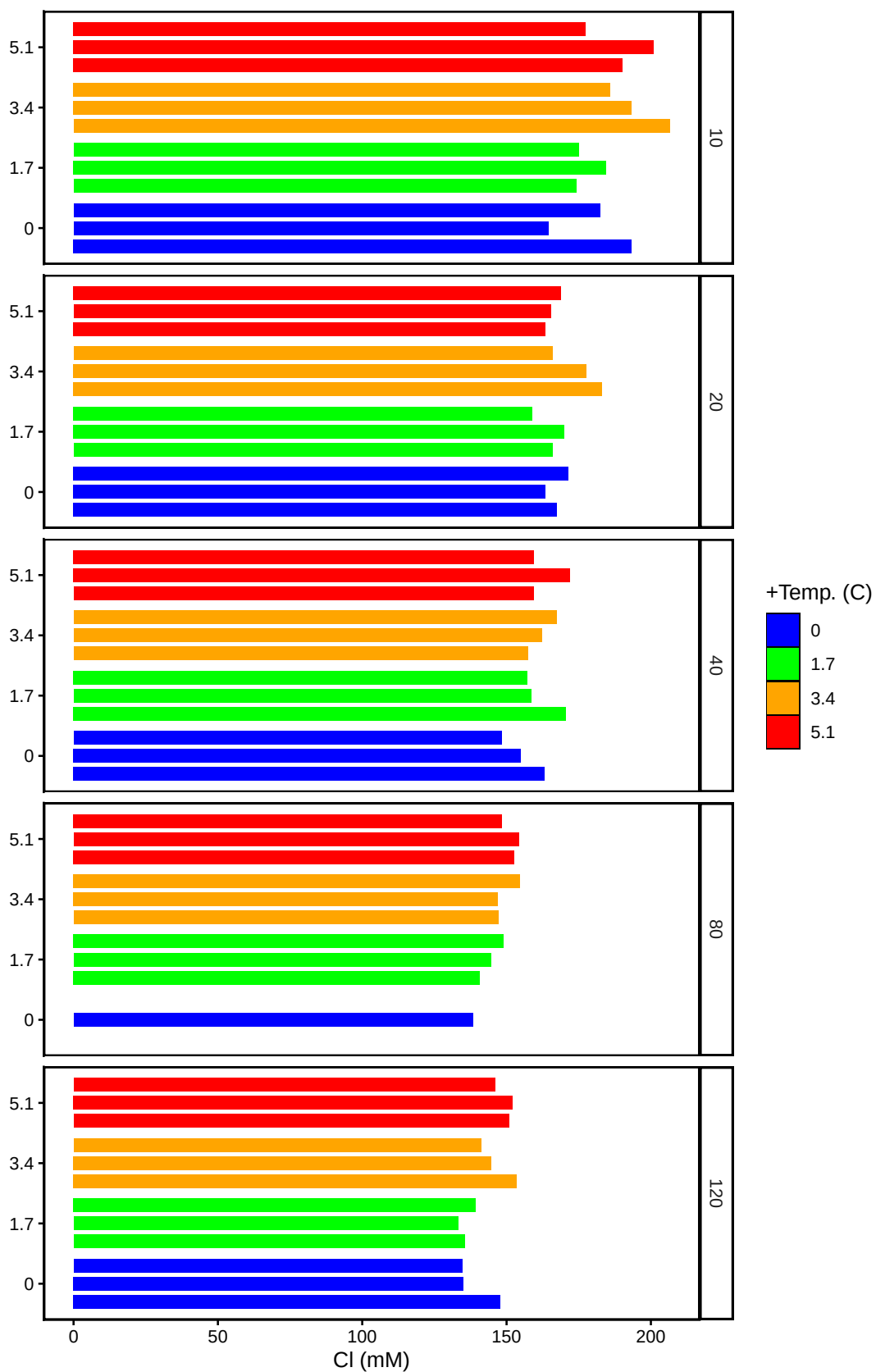
0.10 Visualize Data - Cl

```
## Visualize Data - Cl
```

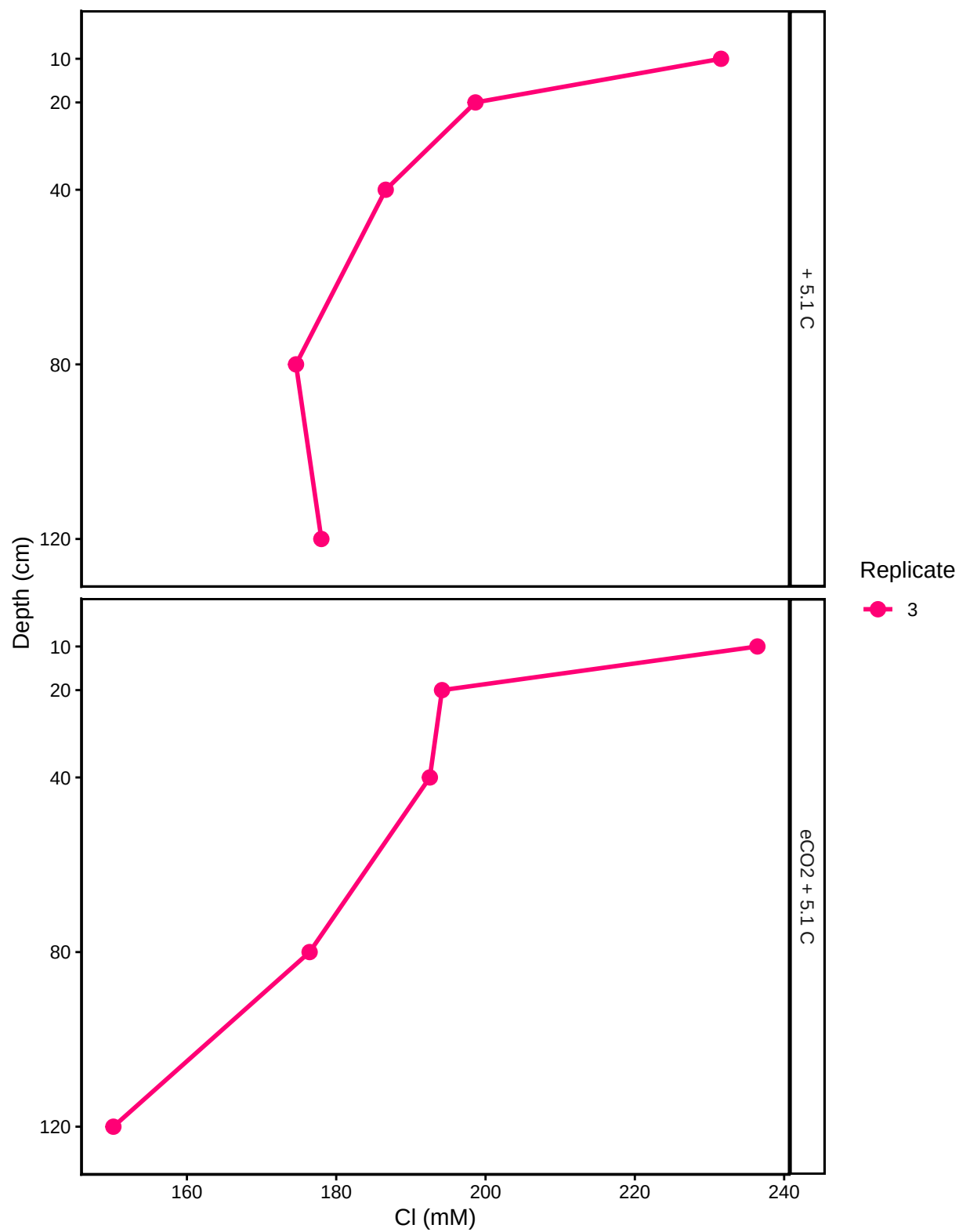
C3: Porewater Cl



C4: Porewater Cl

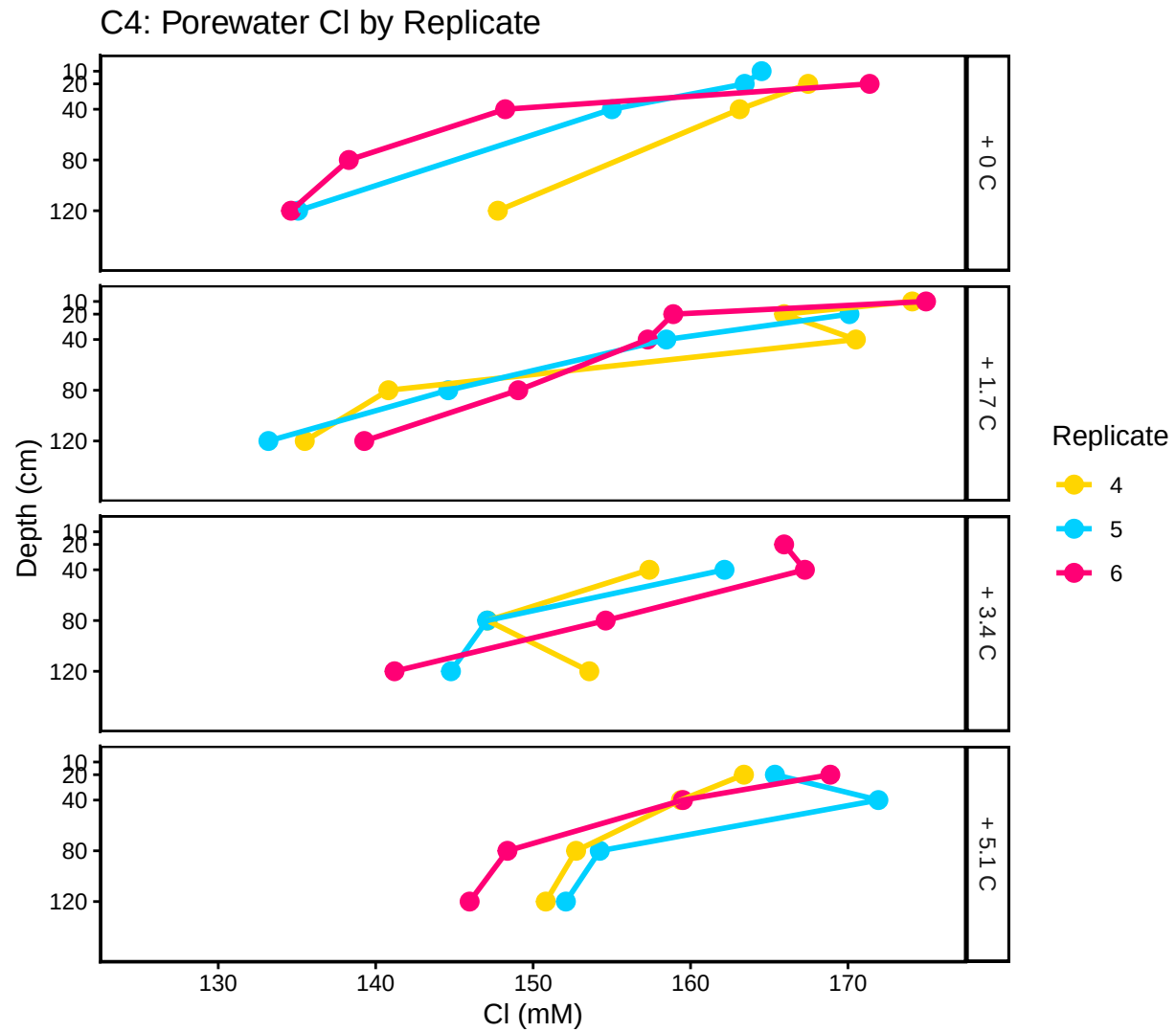


C3: Porewater Cl by Replicate



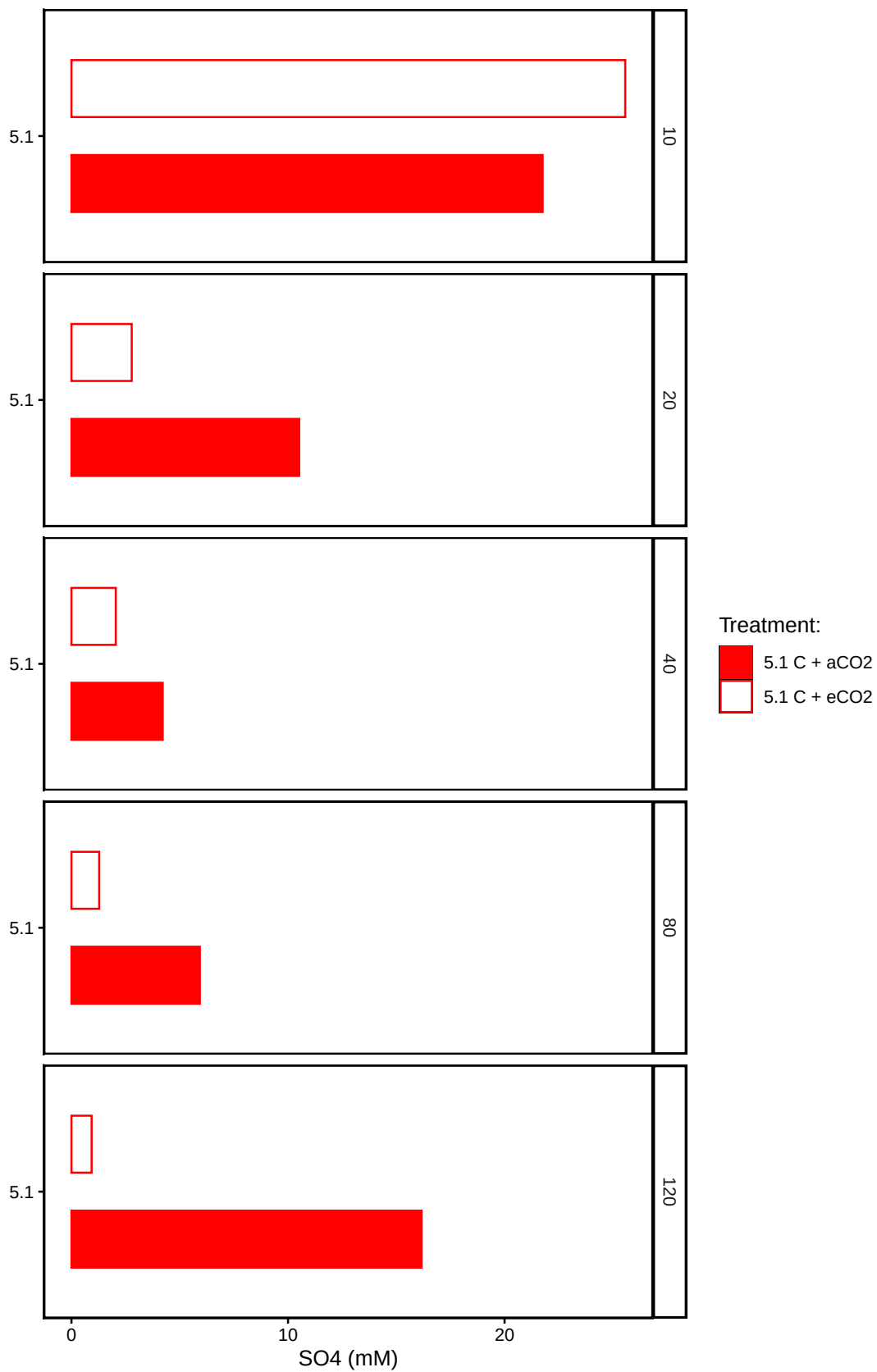
```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_path()').
```

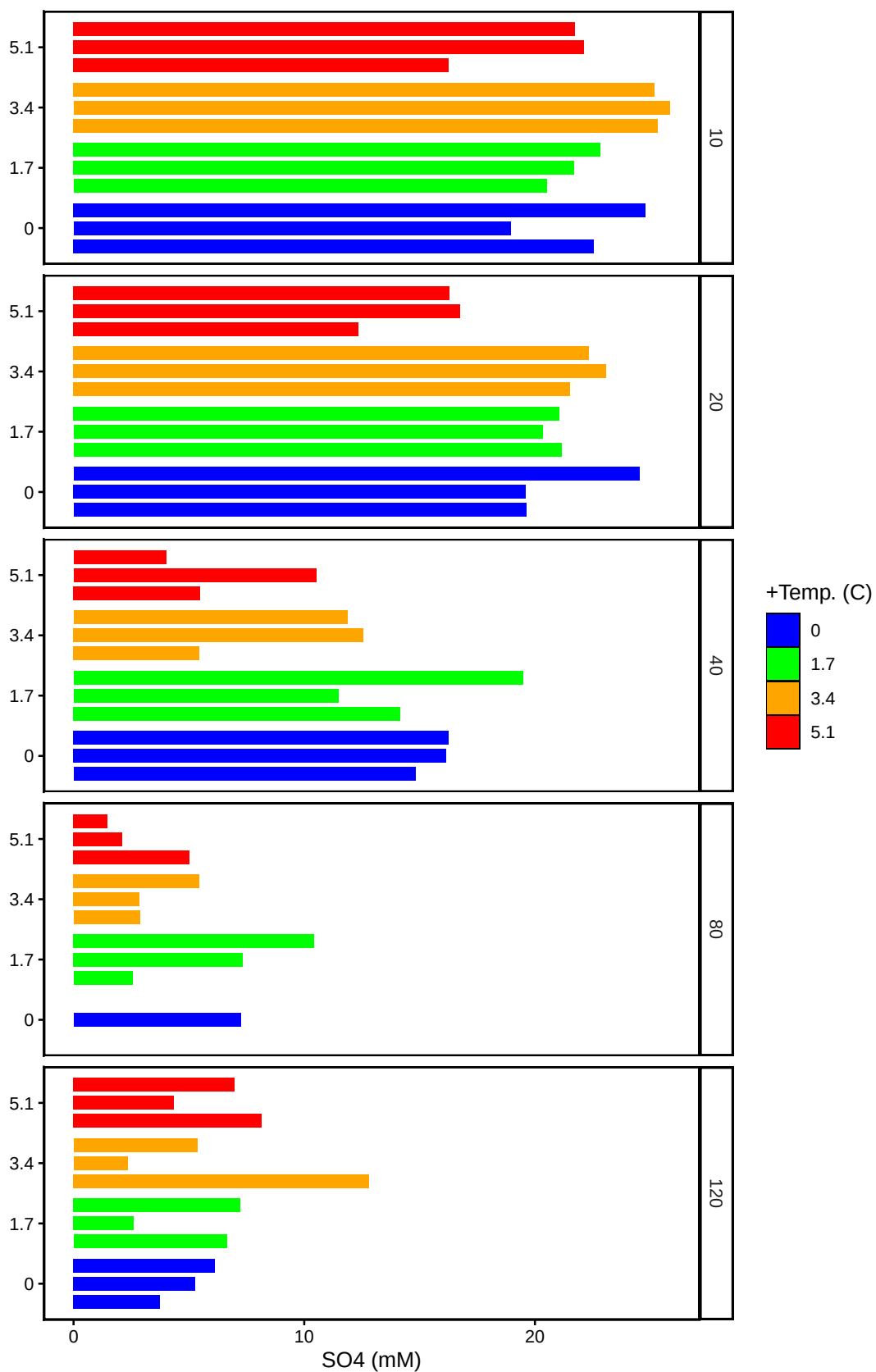


```
## Visualize Data - S04
```

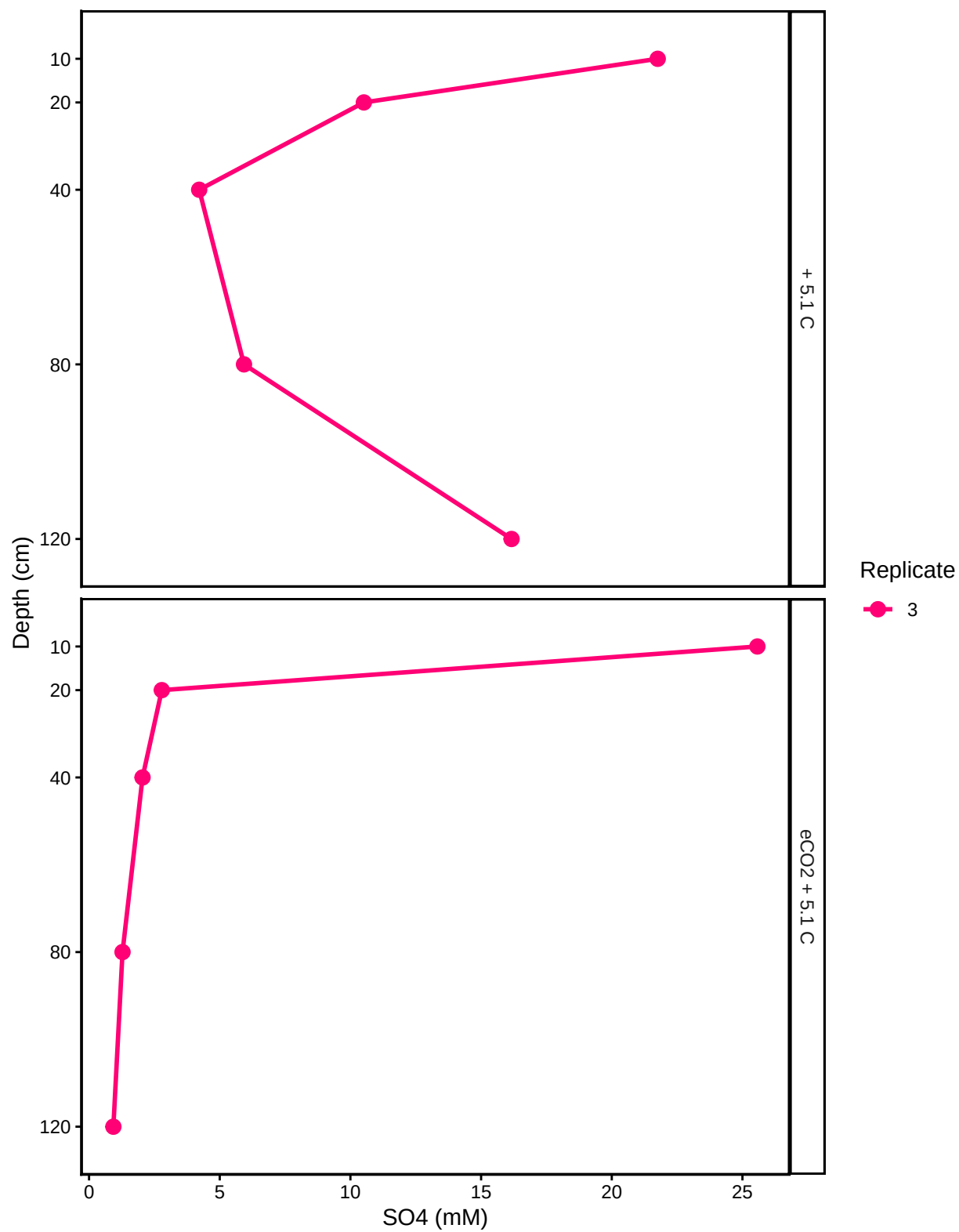
C3: Porewater SO₄

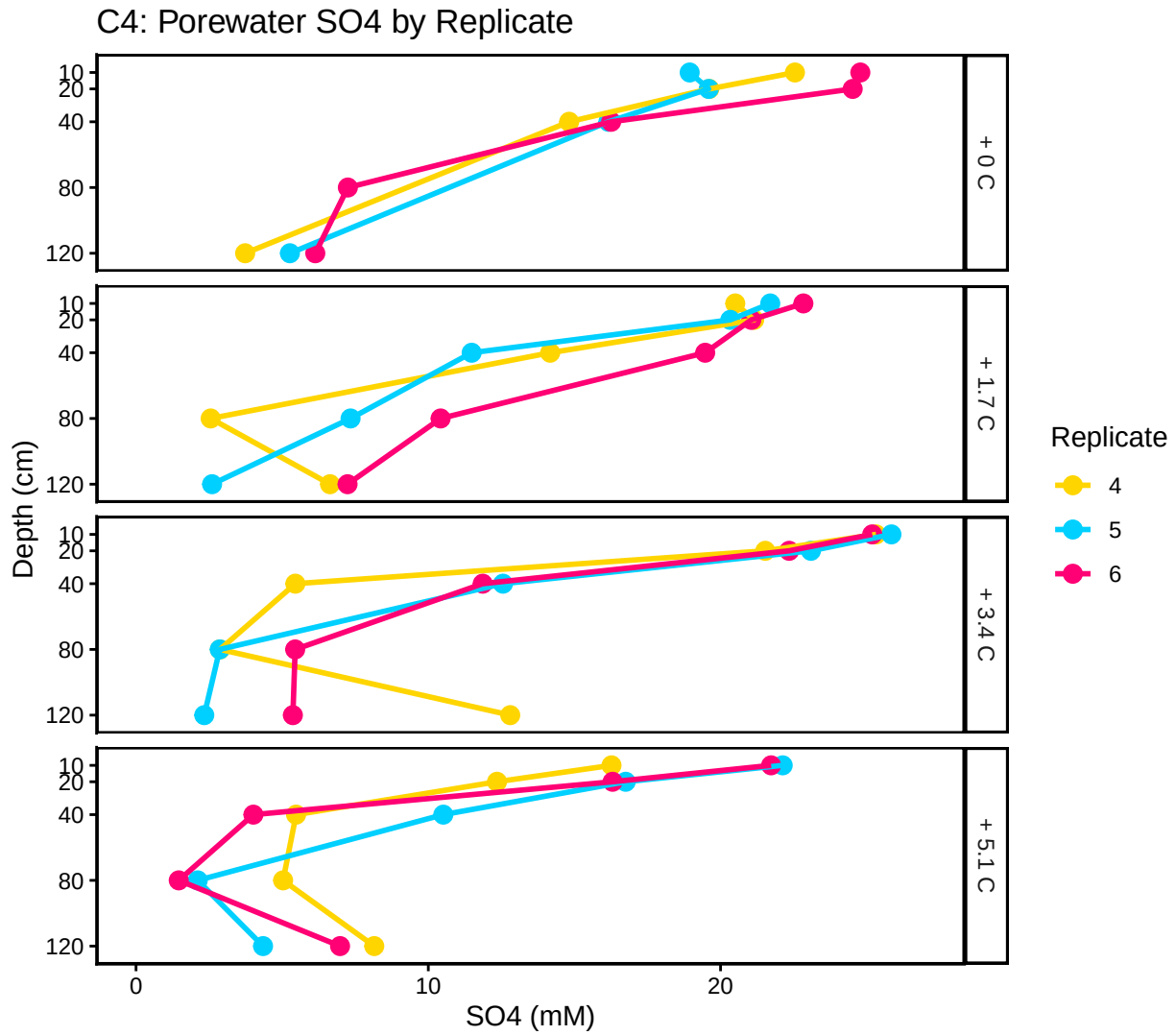


C4: Porewater SO4



C3: Porewater SO₄ by Replicate





0.11 Export Processed Data

#end